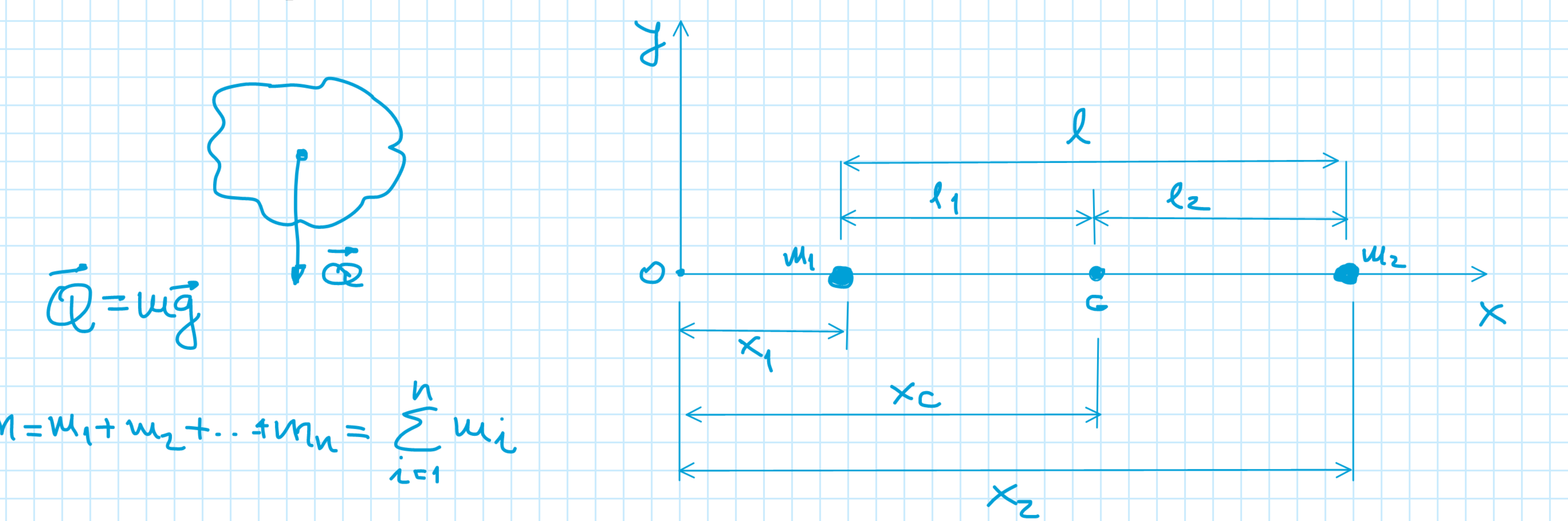


Ligjerata 5b

Qendra e masës dhe qendra rëndesës



$$m=m_1+m_2+...+m_n=\sum_{i=1}^n m_i$$

$$m_1gl_1=m_2gl_2 \quad / :g$$

$$C: Q=(m_1+m_2)g$$

$$m_1l_1=m_2l_2$$

$$l_1=x_c-x_1$$

$$l_2=x_2-x_c$$

$$m_1(x_c-x_1)=m_2(x_2-x_c)$$

$$m_1x_c-m_1x_1=m_2x_2-m_2x_c$$

$$m_1x_c+m_2x_c=m_1x_1+m_2x_2$$

$$x_c(m_1+m_2)=m_1x_1+m_2x_2 \quad / :m_1+m_2$$

$$x_c=\frac{m_1x_1+m_2x_2}{m_1+m_2}$$

$$x_c=\frac{m_1x_1+m_2x_2+...+m_nx_n}{m_1+m_2+...+m_n}=\frac{\sum_{i=1}^n m_ix_i}{\sum_{i=1}^n m_i}$$

$$y_c=\frac{m_1y_1+m_2y_2+...+m_ny_n}{m_1+m_2+...+m_n}=\frac{\sum_{i=1}^n m_iy_i}{\sum_{i=1}^n m_i}$$

$$z_c=\frac{m_1z_1+m_2z_2+...+m_nz_n}{m_1+m_2+...+m_n}=\frac{\sum_{i=1}^n m_iz_i}{\sum_{i=1}^n m_i}$$

→ Qendra e masës ka koordinatat $C(x_c, y_c, z_c)$

$$\vec{C}=x_c\vec{i}+y_c\vec{j}+z_c\vec{k}=\frac{\sum_{i=1}^n m_i\vec{C}_i}{\sum_{i=1}^n m_i}$$